

Kazi Sher Ahmed

Mechatronics Researcher

Karachi, Pakistan

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RESEARCH INTERESTS	Control systems, active magnetic bearings (AMB), rotordynamics, ultra-high-speed motors & spindles	
EDUCATION	Doctoral Studies – Sabanci University, Turkiye Mechatronics Engineering (coursework completed) • Research Topic: Optimization and Realization of AMB Spindle for Chatter Suppression • Advisor: Dr. Bekir Bediz • Relevant Coursework: Machine Learning, Metal Cutting Mechanics, Tribology, Scaling in Energy Systems • Status: Paused in 2023 following a family bereavement	Sep. 2021 – July 2023
	MS – Sabanci University, Turkiye Mechatronics Engineering • Thesis: Design of Active Magnetic Bearing Spindles for Micro-Milling Applications • Advisor: Dr. Bekir Bediz • Relevant Coursework: Real-Time Systems, Integrated Sensors, Mechanical Vibrations, Topology Optimization, Finite Element (FE) Method, Machine Tool Engineering	Sep. 2018 – July 2021
	BS – GIK Institute, Pakistan Mechanical Engineering • Senior Year Project: Development & Finite Element Modeling of a Rotor-Bearing Test Rig • Award: Second Best Senior Year Project	Aug. 2011 – June 2015
MANUSCRIPT UNDER REVIEW	K S Ahmed & B Bediz, 2026, ‘ Design framework and manufacturing of an active magnetic bearing spindle for micro-milling applications ’, arXiv:2603.00169. Revision submitted to <i>The International Journal of Advanced Manufacturing Technology</i> , Aug. 2026	
JOURNAL PAPERS	K S Ahmed , G Kiziltas, & B Bediz, 2025, ‘ Integrated design optimization of a motorized active magnetic bearing spindle for micro-milling applications ’, <i>Journal of Mechanical Science and Technology</i> , vol. 39, pp. 2015–2027	
	K S Ahmed & S M Ahmad, 2019, ‘ VibronRotor, an open-source rotordynamic code: Development and benchmarking ’, <i>Measurement</i> , vol. 131, pp. 546–558	
CONFERENCE PAPERS	K S Ahmed & B Bediz, 2022, ‘ Design of an active magnetic bearing spindle for micro-milling applications ’, <i>19th International Conference on Machine Design and Production</i> , Cappadocia, Turkiye, Aug. 31–Sep. 3	
	A Rehman, K S Ahmed , F A Umrani, B Munir, A Mehboob, S M Ahmad, & Z Kazmi, 2015, ‘ Finite element modelling of a generic rotor-bearing system and experimental validation ’, <i>ASME Dynamic Systems and Control Conference</i> , Ohio, USA, Oct. 28–30	
RESEARCH PROJECTS	Design Framework & Manufacturing of AMB Spindle Graduate Researcher [Preprint] UMTIK Paper	Sabanci University 2019 – 2023 • Proposed an iterative framework to design and manufacture micro-milling AMB spindles, aiming to consolidate fragmented design knowledge • Demonstrated the framework by prototyping a 5-DOF spindle designed for 110,000 rpm with permanent-magnet-biased homopolar AMBs and air turbine drive. Corrected analytical model agreed with 3D electromagnetic finite element (FE) model within 1.17% • Built the prototype through CNC milling, cylindrical grinding, and wire EDM. Validated structural FE models through experimental modal testing

Integrated Design Optimization of Motorized AMB Spindle

Sabancı University

Graduate Researcher [\[JMST Paper\]](#)

2022 – 2023

- Co-wrote a successful grant proposal (TUBITAK Grant: 123M121)
- Developed a multidisciplinary optimization framework integrating AMB and motor performance, conventionally treated separately, for a 5-DOF miniaturized spindle
- Employed analytical models covering electromagnetics, mechanics, & rotordynamics
- Constructed a 3D finite-element model to: (i) analyze and reduce AMB-motor electromagnetic coupling, (ii) numerically correct the AMB analytical model
- Generated Pareto-optimal designs trading torque-to-loss ratio against spindle mass, validated against single-objective optimization at extreme points (maximum deviation 4.1%)

Open-Source Rotordynamic Software

GIK Institute

Laboratory Engineer [\[Measurement Paper\]](#) | [\[Code\]](#)

2017 – 2018

- Developed and benchmarked VibronRotor, an open-source finite-element rotordynamic code compatible with MATLAB and the free scientific programming language GNU Octave
- Implemented key analyses: Campbell diagram, critical speed map, mode shapes, imbalance response, orbit plots, & instability threshold analysis
- Instrumented a rotor test rig with eddy current probes for experimental modal analysis for code validation

INDEPENDENT PROJECT

Open-Source DC Motor Control Test Rig

[\[Code\]](#)

2026

- Built a low-cost STM32-based rig for DC motor characterization and embedded control
- Identified motor parameters from locked-rotor, steady-state, and coast-down tests. Validated the model against the measured position response to a voltage step ($R^2 = 0.998$)
- Implemented PI speed control, pole-placement and LQR position control, and Luenberger/Kalman state estimation. Compared four position-control configurations on matched 0.20 rad steps
- Tested an aggressive LQR tuning with encoder-derived and Kalman-estimated speed feedback. Both exhibited persistent oscillation at the duty limits, so observer-based pole placement was adopted for this rig

SKILLS

Methods: Control system design & embedded implementation, electromagnetics, FE analysis, rotordynamics, design optimization, system identification, experimental modal analysis

Programming & numerical computing: MATLAB, Python, C/C++, Mathematica

CAD & simulation: SolidWorks, COMSOL, ANSYS, MATLAB/Simulink

Experimental & data acquisition: Test rig instrumentation, LabVIEW

PEER REVIEW SERVICE

[Journal of the Brazilian Society of Mechanical Sciences and Engineering, Springer](#)
[Journal of Mechanical Science and Technology, Springer](#)

WORK AND TEACHING EXPERIENCE

Graduate Researcher & Teaching Assistant, Sabancı University Sep. 2018 – May 2023

- Research outcome: 1 journal article (JMST 2025), 1 conference paper (UMTIK 2022), 1 manuscript under review (IJAMT)
- Weekly recitations, grading, and office hours for courses: Control Systems, Mechanical Vibrations, Linear Algebra, Statics, Dynamics, Mechanical Systems, Discrete Mathematics, Calculus
- Mentored undergraduate students on the development of an active magnetic damper

Laboratory Engineer, GIK Institute

Sep. 2016 – Aug. 2018

- Research outcome: 1 journal article (Measurement 2019), open-source rotordynamic code
- Designed & conducted lab experiments in Mechanical Vibrations and Fluid Mechanics
- Teaching assistant course assignments: Fluid Mechanics, Mechanical Vibrations
- Served as Problem-Based Learning Committee Coordinator

Project Engineer, GIK Institute

Sep. 2015 – Aug. 2016

- Designed & conducted computational fluid dynamics analysis of a micro-hydro Kaplan turbine for a Khyber-Pakhtunkhwa Govt. Project
- Delivered a seminar talk on ‘Indigenous R&D of Micro-Hydro Power Plants’, CIIT Lahore (2016)

Summer Intern, K-Electric Combined Cycle Power Plant

June 2014 – July 2014

- Completed on-site trainings on plant operations and delivered technical reports

Research Intern, MARC, Koc University

July 2013 – Aug. 2013

- Designed & conducted stress analysis of a dynamically-loaded spring for linear compressors

HammerHead Arc (Shell Eco-marathon team)

- Team manager (2014–15), Team member (2013–14)
- Led the team and represented Pakistan in Shell Eco-marathon 2015, Philippines
- Designed, fabricated, & tested an ultra energy-efficient electric vehicle

Managing Editor, Science Aurora Magazine

Aug. 2012 – May 2013

- Researched content, wrote articles, & managed the project

AWARDS

- Graduate studies tuition waiver & stipend support at Sabanci University, Turkiye (2018–2023)
- Second position in Senior Year Project Competition at GIK Institute (2015)
- Represented Pakistan in Shell Eco-marathon Asia, Manila, Philippines (2015)
- Best Design Award in National Shell Eco-marathon Off-track Event, Pakistan (2015)
- Sixth position in Battery Electric Category, 2014 Shell Eco-marathon Asia, Manila. Project also won Best Ecological Footprint Award at National Shell Eco-marathon event (2014)

MEMBERSHIP American Society of Mechanical Engineers (ASME)**TEST SCORES** TOEFL: 106/120 (2021)**REFERENCES** **Dr. Bekir Bediz**
Associate Professor, Sabanci University, Turkiye
bbediz@sabanciuniv.edu (MS-PhD advisor)**Dr. Gullu Kiziltas Sendur**
Professor, Sabanci University, Turkiye
gkiziltas@sabanciuniv.edu (Research collaborator)**Dr. Sarvat Mushtaq Ahmad**
Associate Professor, King Fahd University of Petroleum and Minerals, Saudi Arabia
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